Cairo University
Faculty of Engineering

Master & 2nd Year Diploma

Structural Stability – STR 655

May 2012 **Final Exam** Time Allowed = 2 Hours

Answer all questions: OPEN BOOK EXAM

Question no. 1 (30%):

The rigid bar-elastic spring system illustrated in Figure 1 is hinged at A and supported by a roller support at C. The system is provided by rotational spring at A and B with stiffness K_1, and translation spring at B with stiffness K_2, where $K_2 = K_1/L^2$. It is required to compute the following:

1. Elastic buckling load using bifurcation or energy approach.
2. Load-deflection expression after buckling, i.e. post-buckling path.
3. Investigate stability of equilibrium along fundametal path, post-buckling path and at buckling load.

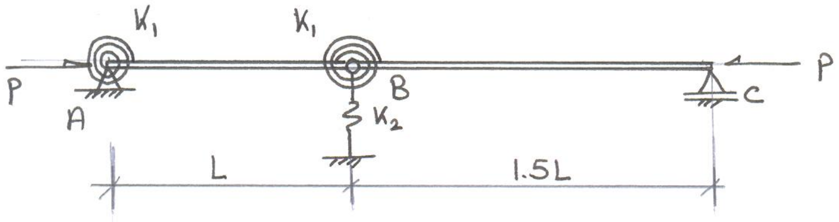

Figure 1: Rigid bar-Elastic Spring System

Question no. 2 (30%):

Figure 2 illustrates a non-sway frame ABCD that is fixed at A and hinged at C & D. It is required to compute the following:

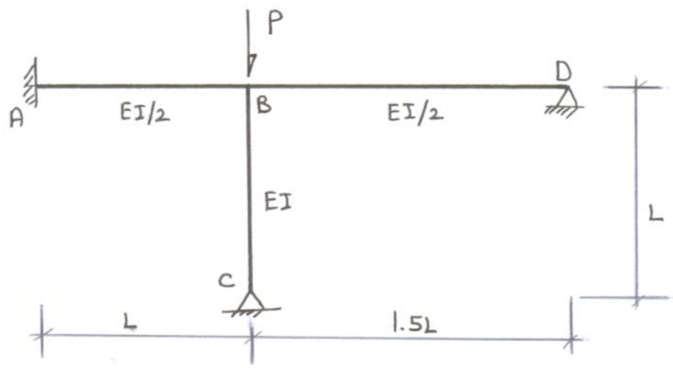

Figure 2: Non-sway frame

a. Buckling load, P_{cr} and effective length factor for column BC.
b. Check your answer in item a using alignment chart. (Hint: Consider G_B and G_C as recommended by ECP-LRFD).
c. If column BC is composed of IPE 400 cross section ($A = 84.5$ cm^2 and $i_x = 16.5$ cm) and knowing that $L = 900$ cm compare ultimate load that can be supported by column BC according to the following ($F_y = 2.4$ t/cm^2):
   1. ECP-LRFD
   2. Inelastic column theory assuming idealized I-section with linear distribution of residual stresses with $F_r = 0.3F_y$
      (Hint: Consider strong axis bending only)

Question no. 3 (30%):

Figure 3 shows the beam column AB fixed at A and supported by a roller support at B. It is required to compute the following:

a. Magnitude and location of M_{max}
b. Equivalent moment factor expressed in the form, $C_m = 1 + \psi\ (P/P_{ek})$
c. Compare C_m computed in item b with that recommended by ECP-LRFD.
d. Compare moment amplification factor stipulated by ECP-LRFD to that obtained by second order analysis at $P/P_{eK} = 0.2$.

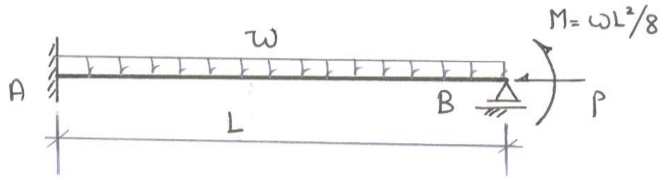

Figure 3: Beam-Column AB

Question no. 4 (30%):

For the sway frame shown in Figure 4 compute the elastic buckling load, P_{cr} and effective length factor, K of column BD if the frame is supported by a roller at A, hinge at D and guided support at C. Check your answer using alignment charts.

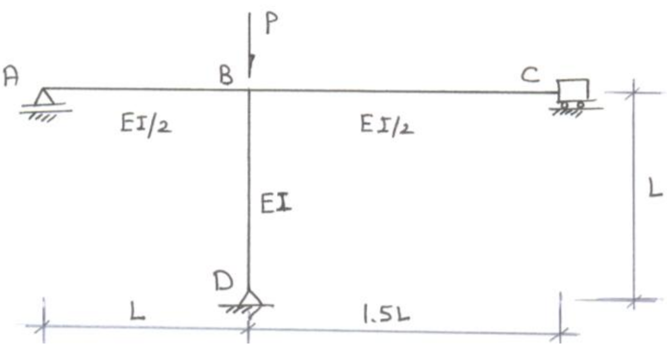

Figure 4: Sway frame ABCD

Good Luck

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